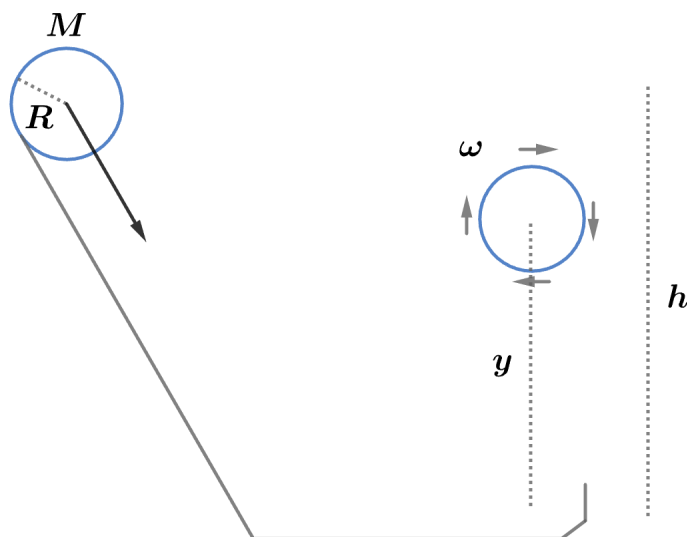


2010 F=ma Exam: Problem 13

Kevin S. Huang



Since the ball rolls without dissipation, we know it is rolling without slipping:

$$v = R\omega$$

Conserving energy between the top and bottom of the ramp,

$$Mgh = K_{\text{trans}} + K_{\text{rot}} = \frac{1}{2}Mv^2 + \frac{1}{2}I\omega^2$$

Using $I = \frac{2}{5}MR^2$ and $v = R\omega$,

$$Mgh = \frac{1}{2}Mv^2 + \frac{1}{2}\left(\frac{2}{5}MR^2\right)\frac{v^2}{R^2} = \frac{7}{10}Mv^2$$

$$v^2 = \frac{10}{7}gh$$

When the ball is launched vertically, only the translational kinetic energy gets converted to gravitational potential energy since the ball keeps rotating:

$$K_{\text{trans}} = \frac{1}{2}Mv^2 = Mgy$$

Using our earlier expression,

$$y = \frac{v^2}{2g} = \frac{5}{7}h$$

so the answer is D.